Laplace Transforms

<u>Laplace Transform</u>: Let *F(t)* be a function of t specified for *t>0*. Then the Laplace Transform of *F(t)*, denoted by $\mathcal{L}\{F(t)\}$ is define by

$\mathcal{L}\{F(t)\} = f(s) = \int_0^\infty e^{-st}F(t)\,dt$, where the parameter s is real.

<u>Some formula of Laplace Transformation</u>

$(i)\ \mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, s > 0$

$(iii)\ \mathcal{L}\{sinat\} = \frac{a}{s^2+a^2}, s > 0$

$(ii)\ \mathcal{L}\{e^{at}\} = \frac{1}{s-a}, s > a$

$(iv)\ \mathcal{L}\{t\} = \frac{1}{s^2}, s > 0$

$(iii)\ \mathcal{L}\{cosat\} = \frac{s}{s^2+a^2}, s > 0$

$(v)\ \mathcal{L}\{1\} = \frac{1}{s}, s > 0$

$(vi)\ \mathcal{L}\{cosh\ at\} = \frac{s}{s^2 - a^2}, s > |a|$

$(vii)\ \mathcal{L}\{sinh\ at\} = \frac{a}{s^2 - a^2}, s > |a|$

<u>Question:</u> Prove that $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, s > a$.

<u>Solution:</u> Let, $F(t) = e^{at}$

By the definition of Laplace transformation, we know that

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st}F(t)\,dt$$

So, $\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st}e^{at}\,dt$

$$= \int_0^\infty e^{-st+at}\,dt = \int_0^\infty e^{-(s-a)t}\,dt$$

$$= \left.\frac{e^{-(s-a)t}}{-(s-a)}\right|_0^\infty = \frac{1}{-(s-a)}\left(e^{-\infty} - e^0\right) = \frac{1}{s-a}, s > a.$$

Question: Prove that $\mathcal{L}\{t\} = \frac{1}{s^2}, s > 0$.

Solution: Let, $F(t) = t$

By the definition of Laplace transformation, we know that

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st} F(t)\, dt$$

So, $\mathcal{L}\{t\} = \int_0^\infty e^{-st} t\, dt$

$$= t\int_0^\infty e^{-st}\, dt - \int_0^\infty \left\{\frac{dt}{dt}\int_0^\infty e^{-st}\, dt\right\} dt$$

$$= t\frac{e^{-st}}{-s}\Bigg|_0^\infty - \int_0^\infty \frac{e^{-st}}{-s} dt = 0 - \frac{1}{-s}.\frac{e^{-st}}{-s}\Bigg|_0^\infty$$

$$= -\frac{1}{s^2}(e^{-\infty} - e^0) = \frac{1}{s^2},\ s > 0.$$

Exercise: $(i)\mathcal{L}\{4e^{at}\} = \frac{4}{s-a}, s > a$

$(ii)\mathcal{L}\{3t\} = \frac{3}{s^2}, s > 0.$

Linear property of Laplace theorem:

If c_1 and c_2 are any constants while $F_1(t)$ and $F_2(t)$ are functions with Laplace transformations $f_1(s)$and $f_2(s)$ respectively, then

$$\mathcal{L}\{c_1F_1(t) + c_2F_2(t)\} = c_1\mathcal{L}\{F_1(t)\} + c_2\mathcal{L}\{F_2(t)\} = c_1f_1(s) + c_2f_2(s)$$

is called the linear property of Laplace Transformations. The result is easily

extended for n terms as follows:

$$\mathcal{L}\{c_1F_1(t) + c_2F_2(t) + \cdots + c_nF_n(t)\} = c_1\mathcal{L}\{F_1(t)\} + c_2\mathcal{L}\{F_2(t)\} + \cdots + c_n\mathcal{L}\{F_n(t)\}.$$

Questions-1: Find the Laplace transformation of $4e^{5t} + 6t^3 - 3\sin 4t + 5\cos 2t$

Solution: By linear property of Laplace transformations, we have

$\mathcal{L}\{4e^{5t} + 6t^3 - 3\sin 4t + 5\cos 2t\}$

$= 4\mathcal{L}\{e^{5t}\} + 6\mathcal{L}\{t^3\} - 3\mathcal{L}\{\sin 4t\} + 5\mathcal{L}\{\cos 2t\}$

$$= 4\left(\frac{1}{s-5}\right) + 6\left(\frac{3!}{s^4}\right) - 3\left(\frac{4}{s^2+4^2}\right) + 5\left(\frac{s}{s^2+2^2}\right)$$

$$= \frac{4}{s-5} + \frac{36}{s^4} - \frac{12}{s^2+16} + \frac{5s}{s^2+4}.$$

Question-2: Find the Laplace transformation of $3t^4 + 4e^{-3t} - 2\sin 5t + 3\cos 2t.$

Solution: By linear property of Laplace transformations, we have

$\mathcal{L}\{3t^4 + 4e^{-3t} - 2\sin 5t + 3\cos 2t\}$

$= 3\mathcal{L}\{t^4\} + 4\mathcal{L}\{e^{-3t}\} - 2\mathcal{L}\{\sin 5t\} + 3\mathcal{L}\{\cos 2t\}$

$$= 3\left(\frac{4!}{s^5}\right) + 4\left(\frac{1}{s+3}\right) - 2\left(\frac{4}{s^2+5^2}\right) + 3\left(\frac{s}{s^2+2^2}\right)$$

$$= \frac{72}{s^5} + \frac{4}{s+3} - \frac{12}{s^2+25} + \frac{3s}{s^2+4}.$$

Exercise: Find the Laplace transformation of the followings:

1. $7t^4 + 5e^{-6t} - 4\sin 5t + 2\cos 2t$
2. $3t^3 + 4e^{-5t} + 3\cos 4t - 2\sin 6t$
3. $10t^6 - 15e^{10t} + 12\sin t + 6\cos 6t$
4. $10\sin 10t - 12t^7 - 2\cos t + 10e^{-t}$
5. $t^3 - 5e^{-7t} - 20\sin 6t$

6. $13e^{10t}+6e^{-t}+12t^8+6\sin t+2\cos 9t$

7. $2e^{-5t}+7e^{6t}+t^7-2\sin 8t+7\cos t$

First translation or shifting property:

If $\mathcal{L}\{\mathrm{F}(t)\}=f(s)$, then $\mathcal{L}\{e^{at}F(t)\}=f(s-a)$

Questions: Find the Laplace transformation of the expression $e^{-2t}(3\cos 6t-5\sin 6t)$.

Solution: We have, $\mathcal{L}\{3\cos 6t-5\sin 6t\}$

$$=3\left(\frac{s}{s^2+6^2}\right)-5\left(\frac{6}{s^2+6^2}\right)$$

$$=\frac{3s}{s^2+36}-\frac{30}{s^2+36}=\frac{3s-30}{s^2+36}$$

Then $\mathcal{L}\{e^{-2t}(3\cos 6t-5\sin 6t)\}=\frac{3(s+2)-30}{(s+2)^2+36}=\frac{3s-24}{s^2+4s+40}$

Exercise: Find the Laplace transformation of the followings:

1. $e^{-4t}(6\sin 3t-5\cos 3t)$
2. $e^{-t}(\cos 2t+2t^2)$
3. $e^{4t}(3t^4+2\sin 7t+\cos 5t)$
4. $e^{-5t}(4t^3+3\cos t-4\sin 6t)$
5. $e^{3t}(7\cos 3t+4\sin 6t)$

The change of scale property:

If $\mathcal{L}\{F(t)\}=f(s)$, then $\mathcal{L}\{F(at)\}\frac{1}{a}f\left(\frac{s}{a}\right)$.

Question: If $\mathcal{L}\{sint\}=\frac{1}{s^2+1}$, Find $\mathcal{L}\{sin3t\}$.

Solution: Here, $\mathcal{L}\{sint\} = \frac{1}{s^2+1}$

Now, $\mathcal{L}\{sin3t\} = \frac{1}{3}\frac{1}{(\frac{s}{3})^2+1} = \frac{3}{s^2+9}$.

Question: If $\mathcal{L}\{f(t)\} = \frac{8(s-3)}{(s^2-6s+25)^2}$, Find $\mathcal{L}\{f(2t)\}$.

Solution: $\mathcal{L}\{f(2t)\} = \frac{1}{2}f\left(\frac{s}{2}\right)$

$$= \frac{1}{2}\left[\frac{8\left(\frac{s}{2}-3\right)}{\left\{\left(\frac{s}{2}\right)^2-6\left(\frac{s}{2}\right)+25\right\}^2}\right]$$

$$= \frac{4\left(\frac{s-6}{2}\right)}{\left(\frac{s^2}{4}-3s+25\right)^2}$$

$$= \frac{2(s-6)}{\left(\frac{s^2-12s+100}{4}\right)^2}$$

$$= \frac{32(s-6)}{(s^2-12s+100)^2}.$$

Question: If $\mathcal{L}\{f(\sqrt{t})\} = \frac{1}{s\sqrt{1+s^2}}$, Find $\mathcal{L}\{f(2\sqrt{t})\}$.

Solution: $\mathcal{L}\{f(\sqrt{t})\} = \frac{1}{4}f\left(\frac{s}{4}\right)$

$$= \frac{1}{4}\left\{\frac{1}{\frac{s}{4}\sqrt{1-\left(\frac{s}{4}\right)^2}}\right\}$$

$$= \frac{1}{4}\left\{\frac{1}{\frac{s}{4}\sqrt{\frac{16+s^2}{16}}}\right\}$$

$$= \frac{1}{4}\frac{16}{s\sqrt{s^2+16}} = \frac{4}{s\sqrt{s^2+16}}.$$

Exercise:

1. If $\mathcal{L}\{cost\} = \frac{1}{s^2+1}$, Then find $\mathcal{L}\{cosat\}$.

2. Suppose $\mathcal{L}\{f(t)\} = \frac{s^2-s+1}{(2s+1)^2(s-1)}$, Then find $\mathcal{L}\{f(2t)\}$.

Laplace Transform of derivatives:

Theorem 1: If $\mathcal{L}\{(t)\} = (s)$, then $\mathcal{L}\{F'(t)\} = sf(s) - F(0)$

Proof: By definition of Laplace transform, we have

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st}F(t)\,dt = f(s)$$

$$\therefore \mathcal{L}\{F'(t)\} = \int_0^\infty e^{-st}F'(t)\,dt$$

$$= e^{-st}\int_0^\infty F'(t)\,dt - \int_0^\infty \left\{\frac{d}{dt}(e^{-st})\int_0^\infty F'(t)dt\right\}dt$$

$$= e^{-st}F(t)\Big|_0^\infty + s\int_0^\infty e^{-st}F(t)\,dt$$

$$= 0 - F(0) + sf(s)$$

$$= sf(s) - F(0).$$

Theorem 2: If $\mathcal{L}\{(t)\} = (s)$, then $\mathcal{L}\{F''(t)\} = s^2f(s) - sF(0) - F'(0)$

Proof: By definition of Laplace transform, we have

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st}F(t)\,dt = f(s)$$

$$\therefore \mathcal{L}\{F''(t)\} = \int_0^\infty e^{-st}F''(t)\,dt$$

$$= e^{-st}\int_0^\infty F''(t)\,dt - \int_0^\infty \left\{\frac{d}{dt}(e^{-st})\int_0^\infty F''(t)dt\right\}dt$$

$$= e^{-st}F'(t)\Big|_0^\infty + s\int_0^\infty e^{-st}F'(t)\,dt$$

$$= 0 - F'(0) + s\int_0^\infty e^{-st}F'(t)\,dt$$

$$= F'(0) + \{sf(s) - F(0)\}$$

$$= s^2(s) - sF(0) - F'(0).$$

Question: If $\mathcal{L}\{(t)\} = \frac{s+1}{s^2+1}$, $F(0) = 1, F'(0) = 0$, Find second derivative of LT.

Solution: We know that

$$\mathcal{L}\{F''(t)\} = s^2f(s) - sF(0) - F'(0)$$

$$= s^2\left(\frac{s+1}{s^2+1}\right) - s.1 - 0$$

$$= s^2\left(\frac{s+1}{s^2+1}\right) - s$$

$$= \frac{s^3+s^2}{s^2+1} - s$$

$$= \frac{s^3+s^2-s^3-s}{s^2+1}$$

$$= \frac{s^2-s}{s^2+1} = \frac{s(s-1)}{s^2+1}.$$

Exercise: (i) If $f(t) = sin5t, F(0) = 0, F'(0) = 5$, Find First and second derivative of LT.

(ii)) If $f(t) = \frac{1}{s-a}, F(0) = 1, F'(0) = 3$, Find First and second derivative of LT.

(iii) If $f(t) = sin2t + cos2t, F(0) = 1, F'(0) = 2$, Find $\mathcal{L}\{F'(t)\}$ and $\mathcal{L}\{F''(t)\}$.

Multiplication by powers of t: Let f (t) be a function and if $\mathcal{L}\{f(t)\} = f(s)$, then $\mathcal{L}\{t^nF(t)\} = (-1)^n\frac{d^n}{ds^n}f(s)$

Question: Find $\mathcal{L}\{t\ sinat\}$.

Solution: Since we know that,

$$\mathcal{L}\{sinat\} = \frac{a}{s^2+a^2}$$

So, $\mathcal{L}\{sinat\} = \left\{(-1)^1 \frac{d}{ds}\left(\frac{a}{s^2+a^2}\right)\right\}$

$$= -\frac{(s^2+a^2).0 - a.2s}{(s^2+a^2)^2}$$

$$= \frac{2as}{s^2+a^2}$$

Question: Find $\mathcal{L}\{t^2 \cos 3t\}$

Solution: Since we know that,

$$\mathcal{L}\{cos3t\} = \frac{s}{s^2+3^2} = \frac{s}{s^2+9}$$

So, $\mathcal{L}\{t^2cos3t\} = \left\{(-1)^2 \frac{d^2}{ds^2}\left(\frac{s}{s^2+9}\right)\right\}$

$$= \frac{d}{ds}\left\{\frac{(s^2+9).1 - s.2s}{(s^2+9)^2}\right\}$$

$$= \frac{d}{ds}\left\{\frac{-s^2+9}{(s^2+9)^2}\right\}$$

$$= \frac{(s^2+9)^2(-2s) - (-s^2+9)2(s^2+9).2s}{(s^2+9)^4}$$

$$= \frac{s(2s^2-54)}{(s^2+9)^3}$$

Question: (i) Find $\mathcal{L}\{t^2e^{2t}\}$

(ii) Find $\mathcal{L}\{t\ cosat\}$

(iii) $\mathcal{L}\{t^2e^{-t}\}$

Inverse Laplace Transform: If $f(s)$ is the Laplace transform of a function $F(t)$, i.e if $\mathcal{L}\{F(t)\} = f(s)$, then $F(t)$ is called the inverse Laplace transform of the function $f(s)$ and is symbolically written as $F(t) = \mathcal{L}^{-1}\{f(s)\}$, where $\mathcal{L}^{-1}$ is called the inverse transformation operator.

Some formula of inverse Laplace Transformation:

$$(\boldsymbol{i})\boldsymbol{\mathcal{L}^{-1}\left\{\frac{1}{s^{n+1}}\right\} = \frac{t^n}{n!}}$$

$$(ii)\boldsymbol{\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}}$$

$$(iii)\boldsymbol{\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} = cosat}$$

$$(iv)\boldsymbol{\mathcal{L}^{-1}\left\{\frac{a}{s^2+a^2}\right\} = sinat}$$

Linearity Property: If $\mathcal{L}\{F_1(t)\} = f_1(s)$ and $\mathcal{L}\{F_2(t)\} = f_2(s)$ and c_1 and c_2 are any two constants, then

$$\mathcal{L}^{-1}\{c_1 f_1(s) + c_2 f_2(s)\} = c_1\mathcal{L}^{-1}\{f_1(s)\} + c_2\mathcal{L}^{-1}\{f_2(s)\} = c_1F_1(t) + c_2F_2(t)$$

Questions-1: Find the inverse Laplace transformation of the expression

$$\frac{1}{s^5} + \frac{4}{s-2} - \frac{3s}{s^2+16} + \frac{5}{s^2+4}$$

Solution: By linear property of inverse Laplace transform,

$$\mathcal{L}^{-1}\left\{\frac{1}{s^5} + \frac{4}{s-2} - \frac{3s}{s^2+16} + \frac{5}{s^2+4}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{s^5}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{s-2}\right\} - 3\mathcal{L}^{-1}\left\{\frac{s}{s^2+16}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{s^2+4}\right\}$$

$$= \frac{1}{24}\mathcal{L}^{-1}\left\{\frac{4!}{s^{4+1}}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - 3\mathcal{L}^{-1}\left\{\frac{s}{s^2+16}\right\} + \frac{5}{2}\mathcal{L}^{-1}\left\{\frac{2}{s^2+4}\right\}$$

$$= \frac{1}{24}t^4 + 4e^{2t} - 3cos4t + \frac{5}{2}sin2t$$

Questions-1: Find the inverse Laplace transformation of the expression

$$\frac{5s+4}{s^5} + \frac{6}{s-3} - \frac{8s}{s^2+9} + \frac{7}{s^2+25}$$

Solution: By linear property of inverse Laplace transform,

$$\mathcal{L}^{-1}\left\{\frac{5s+4}{s^5} + \frac{6}{s-3} - \frac{8s}{s^2+9} + \frac{7}{s^2+25}\right\}$$

$$= \frac{5}{6}\mathcal{L}^{-1}\left\{\frac{3!}{s^{3+1}}\right\} + \frac{4}{24}\mathcal{L}^{-1}\left\{\frac{4!}{s^{4+1}}\right\} + 6\left\{\frac{1}{s-3}\right\} - 8\mathcal{L}^{-1}\left\{\frac{s}{s^2+9}\right\} + \frac{7}{5}\mathcal{L}^{-1}\left\{\frac{5}{s^2+25}\right\}$$

$$= \frac{5}{6}t^3 + \frac{1}{6}t^4 + 6e^{3t} - 8cos3t + \frac{7}{5}sin5t$$

Exercise: Find the Inverse Laplace transformation of the followings:

$(i)\ \frac{6}{s-3} - \frac{3+4s}{s^2+16} + \frac{8-6s}{s^2+9}$

$(ii)\ \frac{1}{s^5} + \frac{4}{s-2} - \frac{3s}{s^2+25} + \frac{5}{s^2+4}$

$(iii)\ \frac{8s+4}{s^6} + \frac{5}{s-1} - \frac{8s}{s^2+36} + \frac{7-5s}{s^2+25}$

First translation or shifting property: If $\mathcal{L}^{-1}\{f(s)\} = F(t)$, then $\mathcal{L}^{-1}\{f(s-a)\} = \mathrm{e}^{at}F(t)$

Question: Find the inverse Laplace transform of the expression $\frac{6s-4}{s^2-4s+20}$

Solution: We have $\mathcal{L}^{-1}\left\{\frac{6s-4}{s^2-4s+20}\right\}$

$$= \mathcal{L}^{-1}\left\{\frac{6s-12+8}{s^2-2.s.2+2^2+16}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{6(s-2)+8}{(s-2)^2+16}\right\}$$

$$= 6\,\mathcal{L}^{-1}\left\{\frac{s-2}{(s-2)^2+16}\right\} + 2\,\mathcal{L}^{-1}\left\{\frac{4}{(s-2)^2+16}\right\}$$

$$= 6e^{2t}cos4t + 2e^{2t}sin4t$$

Question: Find the inverse Laplace transform of the expression $\frac{4s-25}{s^2-6s+34}$

Solution: We have $\mathcal{L}^{-1}\left\{\frac{4s-25}{s^2-6s+34}\right\}$

$$= \mathcal{L}^{-1}\left\{\frac{4s-12-13}{s^2-2.s.3+3^2+25}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{4(s-3)-13}{(s-3)^2+25}\right\}$$

$$= 4\,\mathcal{L}^{-1}\left\{\frac{s-3}{(s-3)^2+5^2}\right\} - \frac{13}{5}\,\mathcal{L}^{-1}\left\{\frac{5}{(s-3)^2+5^2}\right\}$$

$= 4e^{3t}cos5t - \frac{13}{5}e^{3t}sin5t.$

Exercise: $(i)\,\frac{6s-10}{s^2-4s+20}$ $\quad$ $(ii)\,\frac{3s-12}{s^2-6s+18}$

$(iii)\,\frac{10}{s^2+6s+34}$ $\quad$ $(iv)\,\frac{2s+4}{s^2+2s+5}$

Partial Fraction Decomposition:

Question: Evaluate $\mathcal{L}^{-1}\left\{\frac{2s^2-6s+5}{s^3-6s^2+11s-6)}\right\}$

Solution: $\frac{2s^2-6s+5}{s^3-6s^2+11s-6)} = \frac{2s^2-6s+5}{s^3-2s^2-4s^2+8s+3s-6)}$

$$= \frac{2s^2-6s+5}{s^2(s-2)-4s(s-2)+3(s-2)}$$

$$= \frac{2s^2-6s+5}{(s-2)(s^2-4s+3)} = \frac{2s^2-6s+5}{(s-1)(s-2)(s-3)}$$

Let, $\frac{2s^2-6s+5}{(s-1)(s-2)(s-3)} = \frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s-3}$ ……………………………………..(1)

$$\Rightarrow 2s^2-6s+5 = A(s-2)(s-3) + B(s-3)(s-1) + C(s-1)(s-2) \text{ ………………………}(2)$$

Putting s = 1, s = 2, s = 3 in (2), we get

When s = 1, then 2 – 6 + 5 = 2A $\Rightarrow A = \frac{1}{2}$

When s = 2, then 18-12+5 = -B $\Rightarrow B = -1$

When s = 3, then 18 -18 +5=2C $\Rightarrow C = 5/2$

Thus from (1), we have

$$\frac{2s^2-6s+5}{(s-1)(s-2)(s-3)} = \frac{1}{2}\frac{1}{s-1} + \frac{1}{s-2} + \frac{5}{2}\frac{1}{s-3}$$

$$= \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \frac{5}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\}$$

$$= \frac{1}{2}e^{t} - e^{2t} + \frac{5}{2}e^{3t}.$$

Question: Evaluate $\mathcal{L}^{-1}\left\{\frac{5s+3}{(s-1)(s^2+2s+5)}\right\}$

Solution: $\frac{5s+3}{(s-1)(s^2+2s+5)} = \frac{A}{s-1} + \frac{Bs+C}{s^2+2s+5} \dots\dots\dots\dots(1)$

$\Rightarrow 5s+3 = A(s^2+2s+5) + (Bs+C)(s-1) \dots\dots\dots\dots\dots\dots\dots\dots(2)$

$\Rightarrow 5s+3 = As^2 + 2As + 5A + Bs^2 - Bs + Cs - C$

$\Rightarrow 5s+3 = s^2(A+B) + (2A-B+C)s + 5A - C \dots\dots\dots\dots\dots\dots\dots\dots..(3)$

Putting s = 1 in (2), we get

8 =8A $\Rightarrow A = 1$

Equating coefficient of s^2 and s from both sides of (3), we have

$A + B = 0 \Rightarrow B = -1$

$and, \quad 2A - B + C = 5$

$\Rightarrow 2 + 1 + C = 5$

$\Rightarrow C = 2$

Thus from (1), we have

$$\frac{5s+3}{(s-1)(s^2+2s+5)} = \frac{1}{s-1} + \frac{-s+2}{s^2+2s+5}$$

$$= \frac{1}{s-1} + \frac{-s-1+3}{s^2+2s+1+4}$$

$$= \frac{1}{s-1} - \frac{s+1}{(s+1)^2+2^2} + \frac{3}{(s+1)^2+2^2}$$

$$= \frac{1}{2}\,\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \mathcal{L}^{-1}\left\{\frac{s+1}{(s+1)^2+2^2}\right\} + \frac{3}{2}\,\mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2+2^2}\right\}$$

$$= e^t - e^{-t}cos2t + \frac{3}{2}e^{-t}sin2t.$$

Exercise:

1. $\frac{4s-2}{(s^2+1)(s-1)}$

Applications of Laplace transform to Differential Equations

Ordinary differential equations with constant coefiicients:

The Laplace transform is useful in solving linear ordinary differential equations with constant coefficients. For example, suppose we wish to solve the second order linear differential equation

$$\frac{d^2Y}{dt^2} + \alpha \frac{dY}{dt} + \beta Y = F(t)$$

Or,

$$Y'' + \alpha Y' + \beta Y = F(t) \tag{1}$$

Where α and β are constants, subject to the initial or boundary conditions

$$Y(0) = A, \qquad Y'(0) = B \tag{2}$$

Where A and B are constants. On taking the Laplace transform of both sides of (1) and using (2), we obtain an algebraic equation for determination of $\mathcal{L}\{Y(t)\} = y(s)$. The required solution is then obtained by finding the inverse Laplace transform of $y(s)$. The method is easily extended to higher order differential equations.

Mathematical problems:

Question: Solve use of the Laplace transform $Y'' + Y = t,\ Y(0) = 1,\ Y'(0) = -2.$

Solution: Given that, $Y'' + Y = t$

Taking the Laplace transformation on both sides of the differential equation and using the given conditions, we have

$$\mathcal{L}\{Y''\} + \mathcal{L}\{Y\} = \mathcal{L}\{t\},$$

$$s^2 y - sY(0) - Y'(0) + y = \frac{1}{s^2}$$

$$\Rightarrow\ s^2y - s + 2 + y = \frac{1}{s^2}$$

Then

$$y = \mathcal{L}\{Y\} = \frac{1}{s^2(s^2+1)} + \frac{s-2}{s^2+1}$$

$$= \frac{1}{s^2} - \frac{1}{s^2+1} + \frac{s}{s^2+1} - \frac{2}{s^2+1}$$

$$= \frac{1}{s^2} + \frac{s}{s^2+1} - \frac{3}{s^2+1}$$

Again taking inverse Laplace transforms, we have

$$Y = \ \mathcal{L}^{-1}\left\{\frac{1}{s^2} + \frac{s}{s^2+1} - \frac{3}{s^2+1}\right\}$$

$$= t + cost - 3sint.$$

Question: Solve $Y'' + 2Y' + 5Y = e^{-t}sint,\ Y(0) = 0,\ Y'(0) = 1.$

Solution: Given that, $Y'' + 2Y' + 5Y = e^{-t}sint$

Taking the Laplace transformation on both sides of the differential equation and using the given conditions, we have

$$\mathcal{L}\{Y'' + 2Y' + 5Y\} = \ \mathcal{L}\{e^{-t}sint\}$$

$$\Rightarrow \mathcal{L}\left\{Y''\right\} + 2\mathcal{L}\left\{Y'\right\} + 5\mathcal{L}\{Y\} = \mathcal{L}\{e^{-t}sint\}$$

$$\Rightarrow \{s^2y - sY(0) - Y'(0)\} + 2(sy - Y(0)\} + 5y = \frac{1}{(s+1)^2+1}$$

$$\Rightarrow \{s^2y - s(0) - 1\} + 2(sy - 0\} + 5y = \frac{1}{s^2+2s+2}$$

$$\Rightarrow (s^2+2s+5)y - 1 = \frac{1}{s^2+2s+2}$$

$$\Rightarrow y = \frac{1}{s^2+2s+5} + \frac{1}{(s^2+2s+2)(s^2+2s+5)}$$

$$= \frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}$$

Again taking inverse Laplace transforms, we have

$$Y = \mathcal{L}^{-1}\left\{\frac{s^2+2s+3}{(s^2+2s+2)(s^2+2s+5)}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{3(s^2+2s+2)} + \frac{2}{3(s^2+2s+5)}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{1}{3(s^2+2s+2)}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{3(s^2+2s+5)}\right\}$$

$$= \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+1}\right\} + \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{2}{(s+1)^2+4}\right\}$$

$$= \frac{1}{3}e^{-t}sint + \frac{1}{3}e^{-t}sin2t$$

$$= \frac{1}{3}e^{-t}(sint + sin2t).$$

Question: Solve $Y'' - 3Y' + 2Y = 4e^{2t}, Y(0) = -3,\ Y'(0) = 5.$

Solution: Given that, $Y'' - 3Y' + 2Y = 4e^{2t}$

Taking the Laplace transformation on both sides of the differential equation and using the given conditions, we have $\mathcal{L}\left\{Y'' - 3Y' + 2Y\right\} = \mathcal{L}\left\{4e^{2t}\right\}$

$$\Rightarrow \mathcal{L}\left\{Y''\right\} - 3\mathcal{L}\left\{Y'\right\} + 2\mathcal{L}\{Y\} = 4\mathcal{L}\{e^{2t}\}$$

$$\Rightarrow \{s^2y - sY(0) - Y'(0)\} - 3(sy - Y(0)\} + 2y = \frac{4}{s-2}$$

$$\Rightarrow \{s^2y + 3s - 5\} - 3(sy + 3\} + 2y = \frac{4}{s-2}$$

$$\Rightarrow (s^2 - 3s + 2)y + 3s - 14 = \frac{4}{s-2}$$

$$\Rightarrow y = \frac{4}{(s^2-3s+2)(s-2)} + \frac{14-3s}{s^2-3s+2}$$

$$= \frac{-3s^2+20s-24}{(s-1)(s-2)^2}$$

$$= \frac{-7}{s-1} + \frac{4}{s-2} + \frac{4}{(s-2)^2}$$

Again taking inverse Laplace transforms, we have

$$
\begin{aligned}
Y &= \mathcal{L}^{-1}\left\{\frac{-7}{s-1} + \frac{4}{s-2} + \frac{4}{(s-2)^2}\right\} \\
&= \mathcal{L}^{-1}\left\{\frac{-7}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{s-2}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{(s-2)^2}\right\} \\
&= -7e^{t} + 4e^{2t} + 4te^{2t}.
\end{aligned}
$$

Exercise: 1. Solve the initial value problem $Y'' + 2Y' + 5Y = 0, Y(0) = 0,\ Y'(0) = 1.$

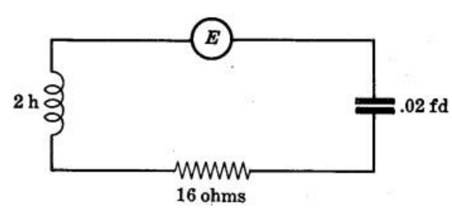
2. Solve the initial value problem using Laplace transform $Y'' + 6Y' + 9Y = e^{-3t}$, $Y(0) = 0,\ Y'(0) = 1.$

3. Solve the initial value problem $Y'' + Y = cos2t, Y(0) = 1,\ Y'(0) = 0.$

Applications to electrical circuits

Question: An inductor of 2 henrys, a resistor of 16 ohms and a capacitor of 0.02 farads are connected in series with an electromotive force (e.m.f) of 300 volts. At $t = 0$ the charge on the capacitor and current in the circuit is zero. Find the charge and current at any time $t > 0$.

Solution: Let Q and I be the instantaneous charge and current respectively at time t.



By Kirchhoff's Laws, we have

$$2\frac{dI}{dt} + 16I + \frac{Q}{0.02} = 300 \quad \ldots\ldots\ldots\ldots \quad (1)$$

Since $I = \frac{dQ}{dt}$, so (1) becomes

$$2\frac{d^2Q}{dt^2} + 16\frac{dQ}{dt} + 50Q = 300$$

$$\Rightarrow \frac{d^2Q}{dt^2} + 8\frac{dQ}{dt} + 25Q = 150 \ \ldots\ldots\ldots\ldots \ (2)$$

With the initial conditions $Q(0) = 0, \ I(0) = Q'(0) = 0$

Taking Laplace transform in (2), we find

$$\mathcal{L}\left\{\frac{d^2Q}{dt^2} + 8\frac{dQ}{dt} + 25Q\right\} = \mathcal{L}\{150\}$$

$$\Rightarrow \{s^2q - sQ(0) - Q'(0)\} + 8\{sq - Q(0)\} + 25q = \frac{150}{s}$$

$$\Rightarrow s^2q + 8sq + 25q = \frac{150}{s}$$

$$\Rightarrow (s^2 + 8s + 25)q = \frac{150}{s}$$

$$\Rightarrow q = \frac{150}{s(s^2 + 8s + 25)}$$

$$\Rightarrow q = \frac{6}{s} - \frac{6s + 48}{s^2 + 8s + 25}$$

$$\Rightarrow q = \frac{6}{s} - \frac{6(s+4) + 24}{(s+4)^2 + 9}$$

$$\therefore q = \frac{6}{s} - \frac{6(s+4)}{(s+4)^2 + 3^2} - \frac{24}{(s+4)^2 + 3^2} \ \ldots\ldots\ldots\ldots \ (3)$$

Taking inverse Laplace transform in (3), we get

$$Q = \mathcal{L}^{-1}\left\{\frac{6}{s} - \frac{6(s+4)}{(s+4)^2 + 3^2} - \frac{24}{(s+4)^2 + 3^2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{6}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{6(s+4)}{(s+4)^2 + 3^2}\right\} - \mathcal{L}^{-1}\left\{\frac{24}{(s+4)^2 + 3^2}\right\}$$

$$= 6 - 6e^{-4t}cos3t - 8e^{-4t}sin3t$$

The current of the circuit is

$$I = \frac{dQ}{dt} = \frac{d}{dt}\{6 - 6e^{-4t}cos3t - 8e^{-4t}sin3t\}$$

$$= 24e^{-4t}cos3t + 32\, e^{-4t}sin3t + 18e^{-4t}sin3t - 24e^{-4t}cos3t$$

$$= 50e^{-4t}sin3t.$$

Question: An inductor of 3 henrys, a resistor of 30 ohms and a capacitor of 0.05 farads are connected in series with an electromotive force (e.m.f) of 150 volts. At $t = 0$ the charge on the capacitor and current in the circuit is zero. Find the charge and current at any time $t > 0$.

Solution: Let Q and I be the instantaneous charge and current respectively at time t

By Kirchhoff's Laws, we have

$$3\frac{dI}{dt} + 30I + \frac{Q}{0.05} = 150 \quad \dots\dots\dots\dots \quad (1)$$

Since $I = \frac{dQ}{dt}$, so (1) becomes

$$3\frac{d^2Q}{dt^2} + 30\frac{dQ}{dt} + 20Q = 150$$

$$\Rightarrow \frac{d^2Q}{dt^2} + 10\frac{dQ}{dt} + \frac{20}{3}Q = 50 \quad \dots\dots\dots\dots \quad (2)$$

With the initial conditions $Q(0) = 0,\ \ I(0) = Q'(0) = 0$

Taking Laplace transform in (2), we find

$$\mathcal{L}\left\{\frac{d^2Q}{dt^2} + 10\frac{dQ}{dt} + \frac{20}{3}Q\right\} = \mathcal{L}\{50\}$$

$$\Rightarrow \{s^2q - sQ(0) - Q'(0)\} + 10\{sq - Q(0)\} + \frac{20}{3}q = \frac{50}{s}$$

$$\Rightarrow s^2q + 10sq + \frac{20}{3}q = \frac{50}{s}$$

$$\Rightarrow \left(s^2 + 10s + \frac{20}{3}\right)q = \frac{50}{s}$$

$$\Rightarrow q = \frac{50}{s\left(s^2 + 10s + \frac{20}{3}\right)}$$

Let, $\frac{50}{s\left(s^2+10s+\frac{20}{3}\right)} = \frac{A}{s} + \frac{Bs+c}{s^2+10s+\frac{20}{3}} \dots\dots\dots\dots\dots\dots\dots\dots. (3)$

$$\Rightarrow 50 = A\left(s^2 + 10s + \frac{20}{3}\right) + (Bs + C)s$$

$$\Rightarrow 50 = As^2 + 10As + \frac{20}{3}A + Bs^2 + Cs$$

$$\Rightarrow 50 = (A+B)s^2 + (10A + C)s + \frac{20}{3}A \quad \dots\dots\dots\dots\dots\dots\dots\dots\dots (4)$$

Equiting coefficients of s^2, s and constants we get,

$$A = \frac{15}{2}, \;\; B = -\frac{15}{2} \;\; and \; C = -75$$

So, from equation (3) we get,

$$\Rightarrow q = \frac{15}{2s} + \frac{-\frac{15}{2}s - 75}{s^2 + 10s + \frac{20}{3}}$$

$$\Rightarrow q = \frac{15}{2s} - \frac{\frac{15}{2}(s + 10)}{(s + 5)^2 - \frac{55}{3}}$$

$$\therefore q = \frac{15}{2s} - \frac{15}{2}\frac{(s + 5)}{(s + 5)^2 - \left(\sqrt{\frac{55}{3}}\right)^2} - \frac{75\sqrt{3}}{2\sqrt{55}}\frac{\sqrt{\frac{55}{3}}}{(s + 4)^2 - \left(\sqrt{\frac{55}{3}}\right)^2} \ldots\ldots\ldots\ldots \;\; (3)$$

Taking inverse Laplace transform in (3), we get

$$Q = \mathcal{L}^{-1}\left\{\frac{15}{2s} - \frac{15}{2}\frac{(s + 5)}{(s + 5)^2 - \left(\sqrt{\frac{55}{3}}\right)^2} - \frac{15\sqrt{165}}{22}\frac{\sqrt{\frac{55}{3}}}{(s + 4)^2 - \left(\sqrt{\frac{55}{3}}\right)^2}\right\}$$

$$= \mathcal{L}^{-1}\left\{\frac{15}{2s}\right\} - \frac{15}{2}\mathcal{L}^{-1}\left\{\frac{(s + 5)}{(s + 5)^2 - \left(\sqrt{\frac{55}{3}}\right)^2}\right\} - \frac{15\sqrt{165}}{22}\mathcal{L}^{-1}\left\{\frac{\sqrt{\frac{55}{3}}}{(s + 4)^2 - \left(\sqrt{\frac{55}{3}}\right)^2}\right\}$$

$$= \frac{15}{2} - \frac{15}{2}e^{-5t}cosh\left(\sqrt{\frac{55}{3}}\right)t - \frac{15\sqrt{165}}{22} \;\; e^{-5t}sinh\left(\sqrt{\frac{55}{3}}\right) \; t.$$

The current of the circuit is

$$I = \frac{dQ}{dt}$$

$$= \frac{d}{dt}\left\{\frac{15}{2} - \frac{15}{2}e^{-5t}cosh\left(\sqrt{\frac{55}{3}}\right)t - \frac{15\sqrt{165}}{22} \; e^{-5t}sinh\left(\sqrt{\frac{55}{3}}\right)t\right\}$$

$$= \frac{15}{2}e^{-5t}\left[-5cosh\left(\sqrt{\frac{55}{3}}\right)t + \sqrt{\frac{55}{3}}sinh\left(\sqrt{\frac{55}{3}}\right)t\right] - \frac{15\sqrt{165}}{22} \; e^{-5t}\left[-5sinh\left(\sqrt{\frac{55}{3}}\right)t + \sqrt{\frac{55}{3}}cosh\left(\sqrt{\frac{55}{3}}\right)t\right]$$

Exercise:

1. An inductor of 2 henrys, a resistor of 16 ohms and a capacitor of 0.02 farads are connected in series with an electromotive force (e.m.f) of $100\ sin3t$ volts. At $t = 0$ the charge on the capacitor and current in the circuit is zero. Find the charge and current at any time $t > 0$.